

Product Requirement Document (PRD)

Project Name: NetOps-CRM (RT/RW Net Management System)

Version: 2.0 (Full Stack & Full Feature Edition)

Target Architecture: Monolith (Node.js + Express) & Mobile (Flutter/React Native)

Core Innovation: QR-Code Driven Field Operations, Real-time Network Telemetry, & Automated Billing.

1. System Architecture & Tech Stack Reference

- **Backend:** Node.js (Express.js)
 - **Database Utama:** PostgreSQL
 - **Database Caching:** Redis
 - **Web Admin Dashboard:** React.js / Vue.js (Tailwind CSS)
 - **Mobile App:** Flutter / React Native
 - **Authentication:** JSON Web Token (JWT) dengan Role-Based Access Control (RBAC).
 - **Hardware Interface:** MikroTik API (`routeros-client`), OLT Telnet/SSH.
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2. Database Schema (PostgreSQL DDL Reference)

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-- 0. Table Users/Admins (RBAC)
CREATE TABLE users (
    id SERIAL PRIMARY KEY,
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    username VARCHAR(50) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    role VARCHAR(20) NOT NULL, -- SUPERADMIN, ADMIN_BILLING, TECHNICIAN
    fullname VARCHAR(100) NOT NULL,
    is_active BOOLEAN DEFAULT TRUE
);

-- 1. Table Clients (Pelanggan)
CREATE TABLE clients (
    id VARCHAR(50) PRIMARY KEY,
    qr_token VARCHAR(255) UNIQUE NOT NULL,
    fullname VARCHAR(100) NOT NULL,
    whatsapp VARCHAR(20) NOT NULL,
    address TEXT NOT NULL,
    coordinates POINT,
    mikrotik_profile VARCHAR(50) NOT NULL,
    monthly_fee NUMERIC(10,2) NOT NULL,
    billing_cycle_date INT DEFAULT 1,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. Table ODP (Optical Distribution Pack)
CREATE TABLE odps (
    id VARCHAR(50) PRIMARY KEY,
    qr_token VARCHAR(255) UNIQUE NOT NULL,
    name VARCHAR(100) NOT NULL,
    total_ports INT DEFAULT 8,
    coordinates POINT NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 3. Table Port Assignments
CREATE TABLE port_assignments (
    id SERIAL PRIMARY KEY,
    odp_id VARCHAR(50) REFERENCES odps(id) ON DELETE CASCADE,
    port_number INT NOT NULL,

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        client_id VARCHAR(50) REFERENCES clients(id) ON DELETE SET NULL,
        status VARCHAR(20) DEFAULT 'FREE',
        UNIQUE(odp_id, port_number)
    );

-- 4. Table Device/ONT Inventory
CREATE TABLE ont_devices (
    mac_address VARCHAR(17) PRIMARY KEY,
    serial_number VARCHAR(50) UNIQUE NOT NULL,
    client_id VARCHAR(50) REFERENCES clients(id) ON DELETE SET NULL,
    last_optical_power NUMERIC(4,2),
    updated_at TIMESTAMP
);

-- 5. Table Invoices (Tagihan)
CREATE TABLE invoices (
    id VARCHAR(50) PRIMARY KEY, -- INV-YYYYMM-XXXX
    client_id VARCHAR(50) REFERENCES clients(id),
    amount NUMERIC(10,2) NOT NULL,
    due_date DATE NOT NULL,
    status VARCHAR(20) DEFAULT 'UNPAID', -- UNPAID, PAID, OVERDUE
    payment_method VARCHAR(50),
    paid_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 6. Table Tickets (Gangguan)
CREATE TABLE tickets (
    id SERIAL PRIMARY KEY,
    client_id VARCHAR(50) REFERENCES clients(id),
    title VARCHAR(150) NOT NULL,
    description TEXT,
    status VARCHAR(20) DEFAULT 'OPEN',
    assigned_technician_id INT REFERENCES users(id),
    closing_photo_url VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

3. Module & Core Features Breakdown

Module 1: Auth & Web Admin Dashboard

1. **Login & RBAC:** Menggunakan JWT. Teknisi tidak bisa mengakses menu keuangan, Admin Billing tidak bisa menghapus konfigurasi MikroTik.
2. **Dashboard Overview:** Menampilkan metrik total pelanggan aktif, total ODP, pendapatan, dan tiket gangguan (OPEN).
3. **Client Management:** Form CRUD untuk mendaftarkan pelanggan, mengatur paket, dan generate QR Code.

Module 2: Field Operations & QR System

1. **Scan ODP:** Mendapatkan data port ODP secara instan dan menghubungkan client baru ke port kosong.
2. **Scan ONT/Modem:** Melakukan diagnostik real-time (Ping, cek Rx Power dBm).
3. **Ticketing Execution:** Teknisi menerima notifikasi, menuju lokasi, memperbaiki, dan mengunggah foto perbaikan (RESOLVED).

Module 3: Automated Billing & CRM

1. **Invoice Generator (Cron Job):** H-3 jatuh tempo otomatis kirim WhatsApp Gateway (nominal & link bayar).
2. **Payment Gateway Webhook:** Menerima notifikasi dari Midtrans/Xendit. Otomatis ubah status ke PAID.
3. **Auto-Isolir (MikroTik Sync):** Telat bayar -> IP/MAC masuk Address List isolir. Lunas -> Isolir dicabut instan.

Module 4: Network Orchestration (MikroTik / OLT API)

1. **Bandwidth Management:** Pengaturan Queue/Profile MikroTik dari Web Admin Dashboard.
 2. **Telemetry Scraper:** Cron job mengecek redaman OLT berkala.
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4. REST API Endpoint Specification

- **POST** `/api/v1/auth/login` : Login user/teknisi, mengembalikan JWT Token.
- **GET** `/api/v1/clients` : Menampilkan daftar pelanggan.
- **POST** `/api/v1/invoices/generate` : Trigger manual/cron membuat tagihan bulan ini.
- **POST** `/api/v1/webhooks/payment` : Menerima callback dari Payment Gateway.
- **GET** `/api/v1/scan/odp/:id` : Status ODP & ketersediaan port.
- **POST** `/api/v1/odp/assign-port` : Menyambung port ODP ke client baru.
- **GET** `/api/v1/scan/client/:id/status` : API call real-time ke MikroTik & OLT.
- **POST** `/api/v1/client/reboot-ont` : Remote reboot ONT pelanggan via OLT.
- **PATCH** `/api/v1/tickets/:id/resolve` : Upload bukti foto perbaikan (Multipart).